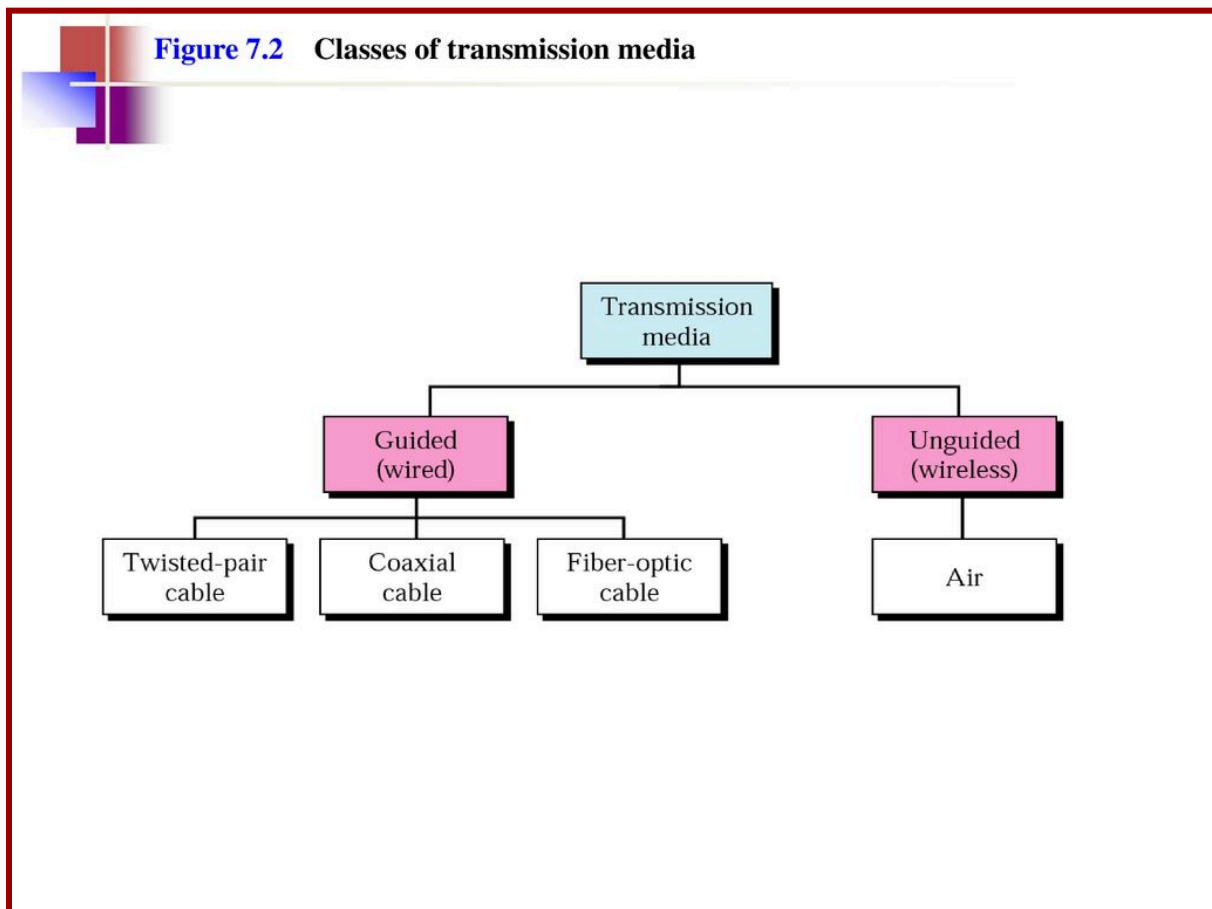


Chapter 3: Transmission Media and Data Compression

3.1 Guided Transmission Media: Twisted Pair, Co-axial Cable, Optical Fiber.

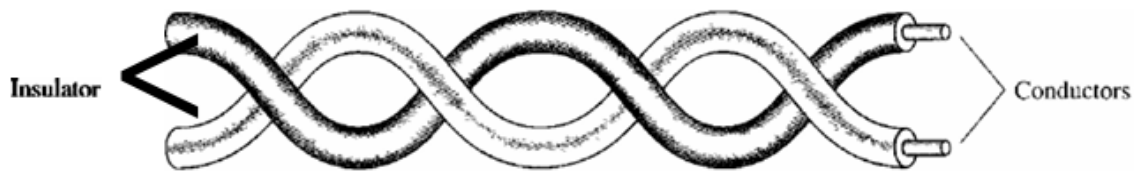
Guided media, which are those that provide a conduit from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable. A signal traveling along any of these media is directed and contained by the physical limits of the medium



Twisted-pair and coaxial cable use metallic (copper) conductors that accept and transport signals in the form of electric current. Optical fiber is a cable that accepts and transports signals in the form of light.

Twisted-Pair Cable:

A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together, as shown in Figure

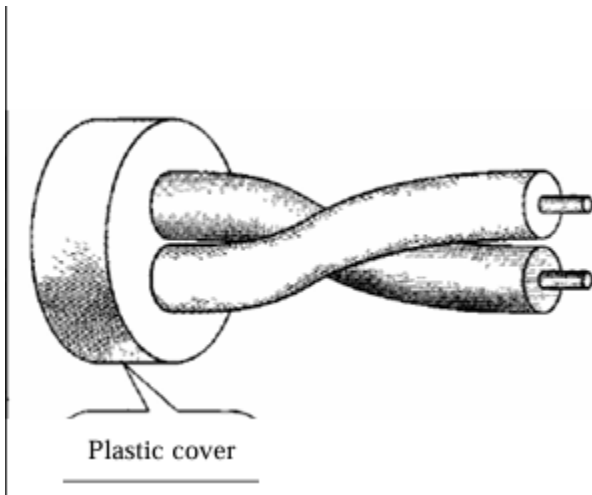


- >One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference.
- >The receiver uses the difference between the two.
- >In addition to the signal sent by the sender on one of the wires, interference (noise) and crosstalk may affect both wires and create unwanted signals.
- >If the two wires are parallel, the effect of these unwanted signals is not the same in both wires because they are at different locations relative to the noise or crosstalk sources (e.g., one is closer and the other is farther).
- >This results in a difference at the receiver. By twisting the pairs, a balance is maintained.
- >For example, suppose in one twist, one wire is closer to the noise source and the other is farther; in the next twist, the reverse is true.
- >Twisting makes it probable that both wires are equally affected by external influences (noise or crosstalk).
- >This means that the receiver, which calculates the difference between the two, receives no unwanted signals.
- >The unwanted signals are mostly canceled out. From the above discussion, it is clear that the number of twists per unit of length (e.g., inch) has some effect on the quality of the cable.

Unshielded Versus Shielded Twisted-Pair Cable

1. Unshielded Twisted Pair (UTP):

UTP consists of two insulated copper wires twisted around one another. This type of cable has the ability to block interference and does not depend on a physical shield for this purpose. It is used for telephonic applications.



Advantages of Unshielded Twisted Pair

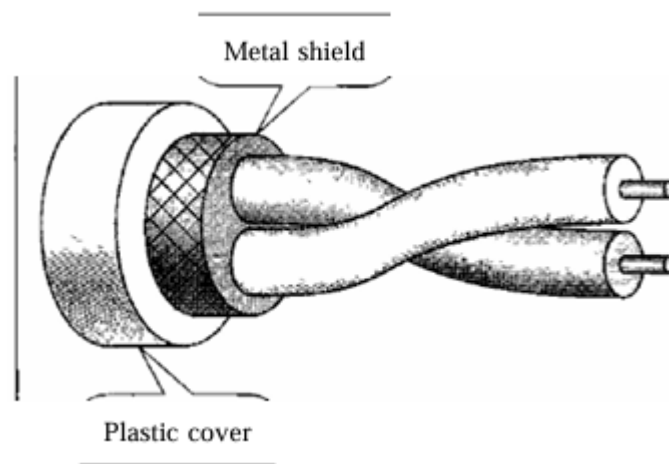
- Least expensive
- Easy to install
- High-speed capacity

Disadvantages of Unshielded Twisted Pair

- Lower capacity and performance in comparison to STP
- Short distance transmission due to attenuation

2. Shielded Twisted Pair (STP)

STP cable consists of a special jacket (a copper braid covering or a foil shield) to block external interference. It is used in fast data rate Ethernet and in voice and data channels of telephone lines.



Advantages of Shielded Twisted Pair

- Better performance at a higher data rate in comparison to UTP
- Eliminates crosstalk
- Comparatively faster

Disadvantages of Shielded Twisted Pair

- Comparatively difficult to install and manufacture
- More expensive
- Bulky

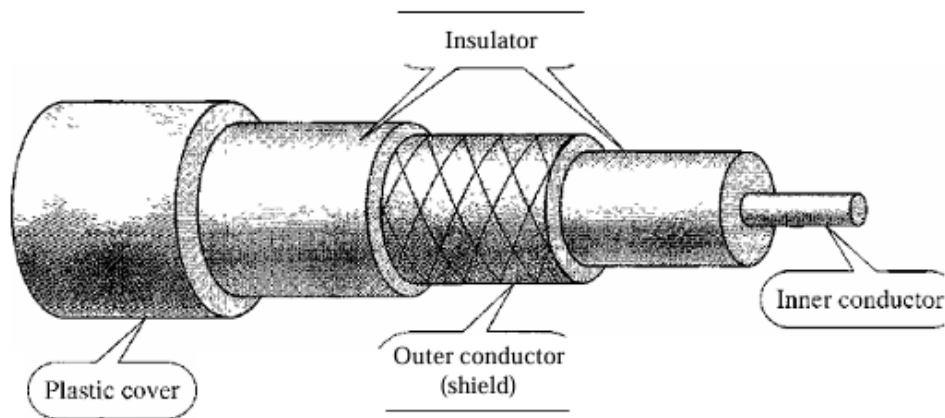
Coaxial Cable

->Coaxial cable (or coax) carries signals of higher frequency ranges than those in twisted pair cable.

->Coaxial cable has an outer plastic covering containing an insulation layer made of PVC or Teflon and 2 parallel conductors each having a separate insulated protection cover.

->The Coaxial cable transmits information in two modes: Baseband mode(dedicated cable bandwidth) and Broadband mode(cable bandwidth is split into separate ranges).

->Cable TVs and analog television networks widely use Coaxial cables.



Advantages of Coaxial Cable

- Coaxial cables have high bandwidth .
- It is easy to install.
- Coaxial cables are more reliable and durable.
- Less affected by noise or crosstalk or electromagnetic interference.
- Coaxial cables support multiple channels

Disadvantages of Coaxial Cable

- Coaxial cables are expensive.
- The coaxial cable must be grounded in order to prevent any crosstalk.
- As a Coaxial cable has multiple layers it is very bulky.
- There is a chance of breaking the coaxial cable and attaching a “t-joint” by hackers, this compromises the security of the data.

Optical Fiber Cable

- >An optical fiber is a thin, flexible medium capable of guiding an optical ray.
- >Various glasses and plastics can be used to make optical fibers.
- >The lowest losses have been obtained using fibers of ultrapure fused silica. ->
- >But Ultrapure fiber is difficult to manufacture
- >Hence higher-loss multicomponent glass fibers are used as more economical and still provide good performance.
- >Plastic fiber is even less costly and can be used for short-haul links, for which moderately high losses are acceptable.

Construction

- >An optical fiber strand (also called an optical waveguide) has a cylindrical shape and consists of three concentric sections: the core, the cladding, and the buffer coating.
- >The core is the innermost section and consists of thin strands made of glass or plastic; the core has a diameter in the range of 8 to 62.5 mm.
- >The core is surrounded by a cladding, which is a glass or plastic coating that has optical properties different from those of the core and a diameter of 125 mm.
- >The interface between the core and cladding acts as a reflector to confine light that would otherwise escape the core.

- >The outermost layer is the buffer coating, which is a hard plastic coating that protects the glass from moisture and physical damage.
- >Fiber optic cable provides protection to the fiber from stress during installation and from the environment once it is installed.
- >Cables may contain from only one to hundreds of fibers inside.
- >The outermost layer of the cable, surrounding one or a bundle of fibers, is the jacket.
- >The jacket is composed of plastic and other materials, layered to protect against moisture, abrasion, crushing, and other environmental dangers.

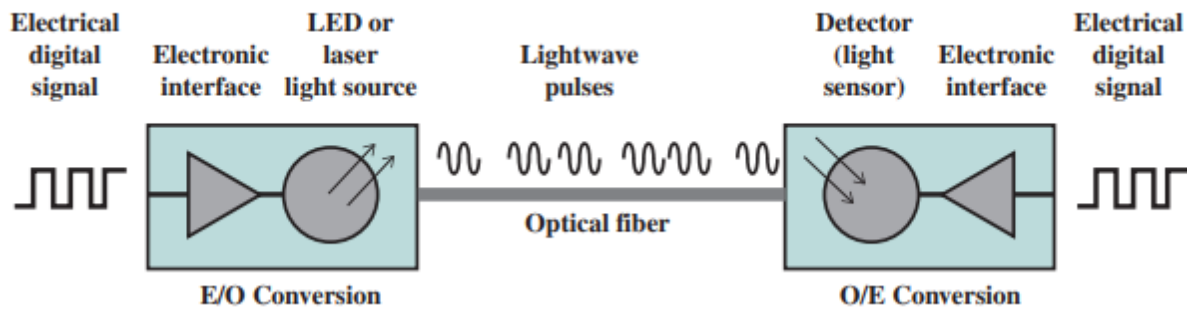


Fig: 1

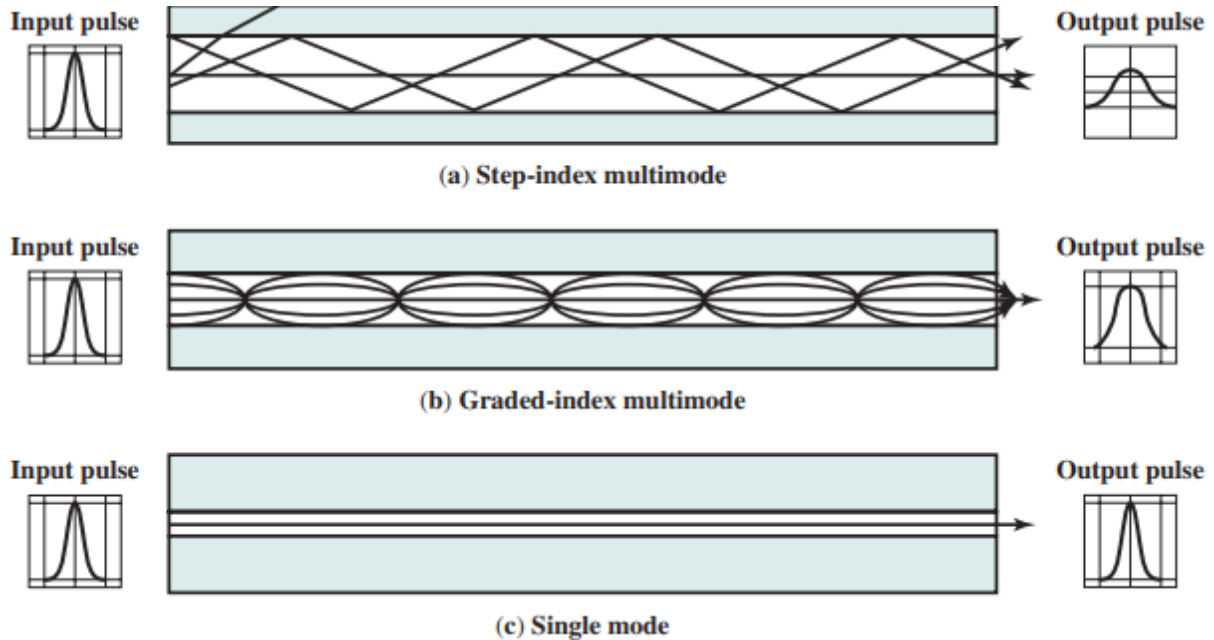


Fig: 2

- >Figure 1 shows the general structure of a fiber optic link, which consists of a transmitter on one end of a fiber and a receiver on the other end.
- >Most systems operate by transmitting in one direction on one fiber and in the reverse direction on another fiber for full duplex operation.
- >The transmitter takes as input a digital electrical signal. This signal feeds into a LED or laser light source through an electronic interface.
- >The light source produces a series of lightwave pulses that encode the digital data from the electrical input.
- >The receiver includes a light sensor that detects the incoming light signal and converts it back to a digital electrical signal.
- >Optical fiber transmits a signal-encoded beam of light by means of total internal reflection.
- >Total internal reflection can occur in any transparent medium that has a higher index of refraction than the surrounding medium.

->Figure 2 shows the principle of optical fiber transmission. Light from a source enters the cylindrical glass or plastic core. Rays at shallow angles are reflected and propagated along the fiber; other rays are absorbed by the surrounding material. This form of propagation is called **step-index multimode**, referring to the variety of angles that reflect.

->With multimode transmission, multiple propagation paths exist, each with a different path length and hence time to traverse the fiber.

->This causes signal elements (light pulses) to spread out in time, which limits the rate at which data can be accurately received. Put another way, the need to leave spacing between the pulses limits data rate.

->This type of fiber is best suited for transmission over very short distances.

->When the fiber core radius is reduced, fewer angles will reflect. By reducing the radius of the core to the order of a wavelength, only a single angle or mode can pass: the axial ray. [Figure 2 c]

->This single-mode propagation provides superior performance for the following reason.

->Because there is a single transmission path with single-mode transmission, the distortion found in multimode cannot occur.

->Single mode is typically used for long-distance applications, including telephone and cable television.

->Finally, by varying the index of refraction of the core, a third type of transmission, known as **graded-index multimode**, is possible.

->This type is intermediate between the other two in characteristics.

->The higher refractive index (discussed subsequently) at the center makes the light rays moving down the axis advance more slowly than those near the cladding.

->Rather than zig-zagging off the cladding, light in the core curves helically because of the graded index, reducing its travel distance.

->The shortened path and higher speed allows light at the periphery to arrive at a receiver at about the same time as the straight rays in the core axis.

-> Graded-index fibers are often used in LANs.

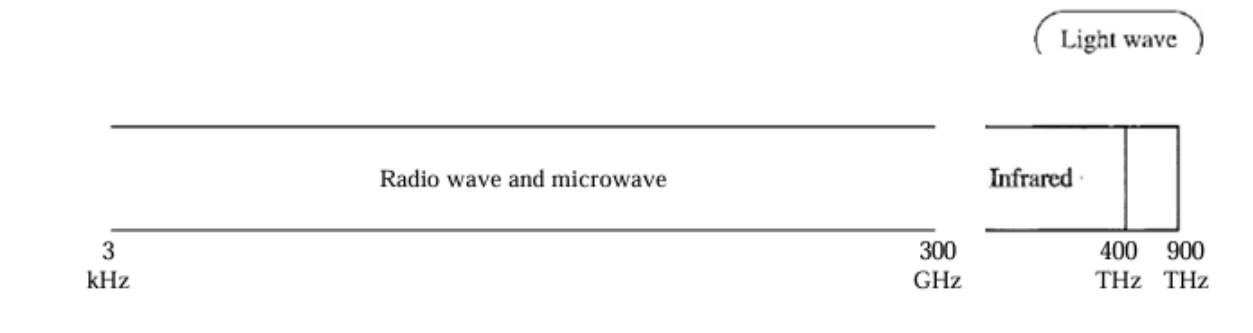
Feature	Single-Mode Fiber	Multimode Fiber
Light Propagation	Single path (one mode)	Multiple paths (multiple modes)
Light Source	Laser diode	LED
Bandwidth	Very high	Lower compared to single-mode
Transmission Distance	Long distances (up to 100+ km)	Short distances (up to 2 km or less)
Modal Dispersion	Minimal or none	Significant due to multiple paths
Cost	Higher (fiber and equipment)	Lower cost
Common Applications	Long-haul communication, WAN, telecom	LAN, data centers, short-range links

Transmission Medium	Typical Frequency Range	Bandwidth	Maximum Data Rate
Twisted Pair	~0 – 600 MHz (Cat 6a)	Up to ~500–600 MHz	Up to 10 Gbps (Cat 6a over ~100m)

Coaxial Cable	$\sim 0 - 3$ GHz (depending on type)	Up to $\sim 1-3$ GHz	Up to 10 Gbps (DOCSIS 3.1 / high-grade coax)
Optical Fiber	$\sim 10^{12} - 10^{15}$ Hz (infrared light)	Tens of THz (very wide)	Tbps+ (with WDM and DWDM tech)

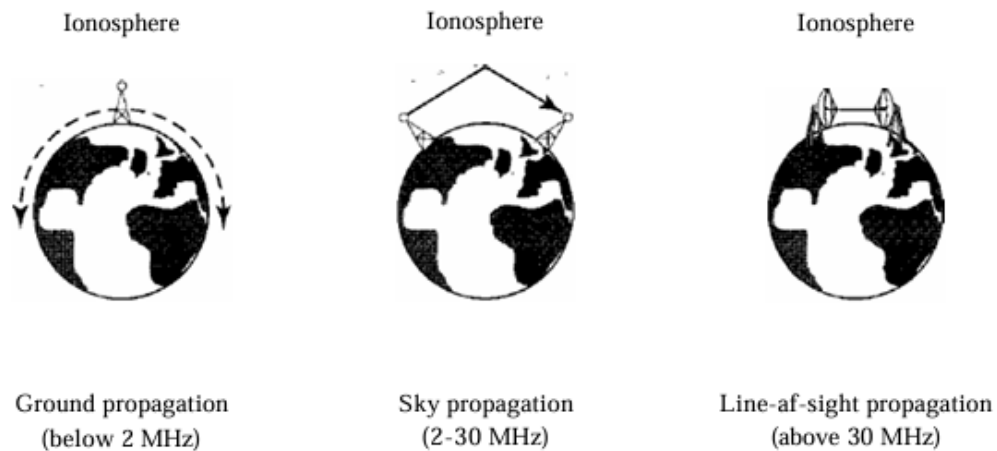
3.2 Unguided Transmission Media: Wireless

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication. Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.



Unguided signals can travel from the source to destination in several ways: ground propagation, sky propagation, and line-of-sight propagation, as shown in Figure. In ground propagation, radio waves travel through the lowest portion of the atmosphere, hugging the earth. These low-frequency signals emanate in all directions from the transmitting antenna and follow the curvature of the planet. Distance depends on the amount of power in the signal: The greater the power, the greater the distance. In sky propagation, higher-frequency radio waves radiate

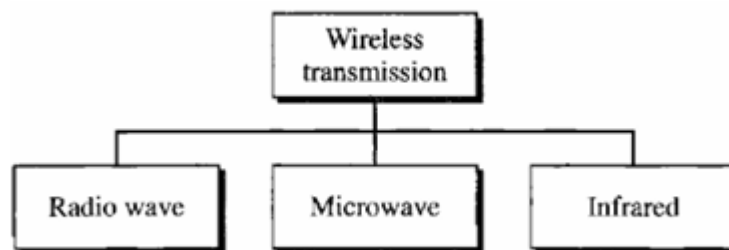
upward into the ionosphere (the layer of atmosphere where particles exist as ions) where they are reflected back to earth. This type of transmission allows for greater distances with lower output power. In line-of-sight propagation, very high-frequency signals are transmitted in straight lines directly from antenna to antenna. Antennas must be directional, facing each other,



and either tall enough or close enough together not to be affected by the curvature of the earth. Line-of-sight propagation is tricky because radio transmissions cannot be completely focused. The section of the electromagnetic spectrum defined as radio waves and microwaves is divided into eight ranges, called bands, each regulated by government authorities. These bands are rated from very low frequency (VLF) to extremely high frequency (EHF).

<i>Band</i>	<i>Range</i>	<i>Propagation</i>	<i>Application</i>
VLF (very low frequency)	3-30 kHz	Ground	Long-range radio navigation
LF (low frequency)	30-300 kHz	Ground	Radio beacons and navigational locators
MF (middle frequency)	300 kHz-3 MHz	Sky	AM radio
HF (high frequency)	3-30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30-300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz-3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3-30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30-300 GHz	Line-of-sight	Radar, satellite

We can divide wireless transmission into three broad groups: **radio waves, microwaves, and infrared waves.**



Radio Waves are easy to generate and can penetrate through buildings.

Radio waves, for the most part, are omnidirectional. When an antenna transmits radio waves, they are propagated in all directions. This means that the sending and receiving antennas do not have to be aligned.

. Frequency Range: 3KHz - 1GHz.

Radio waves use omnidirectional antenna.

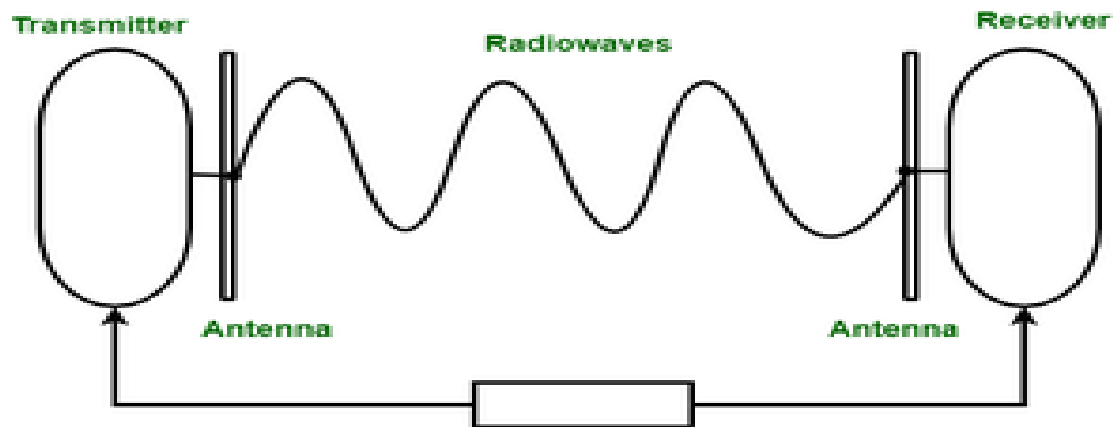
AM and FM radios and cordless phones use Radio waves for transmission.

Types of Radio Waves:

- **Short Wave:** AM Radio
- **VHF (Very High Frequency):** FM Radio/TV
- **UHF (Ultra High Frequency):** TV

Radio Wave Components:

- **Transmitter:** Responsible for encoding the signal.
- **Receiver:** Responsible for decoding the signal.



Microwaves

It is a line of sight transmission i.e. the sending and receiving antennas need to be properly aligned with each other. The distance covered by the signal is directly proportional to the height of the antenna. Frequency Range: 1GHz -

300GHz. **Microwaves** are majorly used for mobile phone communication and television distribution.

Microwaves are unidirectional. When an antenna transmits microwave waves, they can be narrowly focused. This means that the sending and receiving antennas need to be aligned. The unidirectional property has an obvious advantage. A pair of antennas can be aligned without interfering with another pair of aligned antennas. The following describes some characteristics of microwave propagation:

- Microwave propagation is line-of-sight. Since the towers with the mounted antennas need to be in direct sight of each other, towers that are far apart need to be very tall. The curvature of the earth as well as other blocking obstacles do not allow two short towers to communicate by using microwaves. Repeaters are often needed for long-distance communication.
- Very high-frequency microwaves cannot penetrate walls. This characteristic can be a disadvantage if receivers are inside buildings.
- The microwave band is relatively wide, almost 299 GHz. Therefore wider subbands can be assigned, and a high data rate is possible
- Use of certain portions of the band requires permission from authorities.

Advantages:

- Cheaper than using cables
- Freedom from land acquisition
- Ease of communication in difficult terrains
- Communication over oceans

Disadvantages:

- Insecure communication.
- Out of phase signal.
- Susceptible to weather conditions.

- Bandwidth is limited.
- High cost of design, implementation, and maintenance.

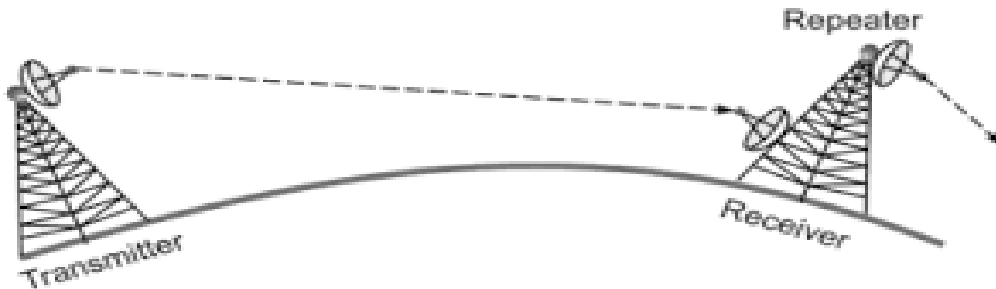


Fig: Microwave Transmission

Infrared[Frequency Range: 300GHz - 400THz]

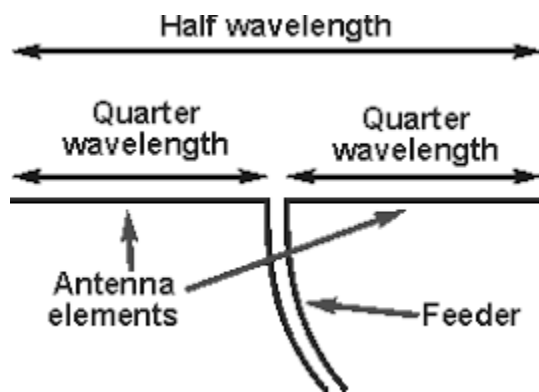
Infrared waves are used for very short distance communication. They cannot penetrate through obstacles. This advantageous characteristic prevents interference between one system and another; a short-range communication system in one room cannot be affected by another system in the next room. When we use our infrared remote control, we do not interfere with the use of the remote by our neighbors. However, this same characteristic makes infrared signals useless for long-range communication. In addition, we cannot use infrared waves outside a building because the sun's rays contain infrared waves that can interfere with the communication. Frequency Range: 300GHz - 400THz. It is used in TV remotes, wireless mouse, keyboard, printer, etc.

The Infrared Data Association (IrDA), an association for sponsoring the use of infrared waves, has established standards for using these signals for communication between devices such as keyboards, mice, PCs, and printers. For example, some manufacturers provide a special port called the IrDA port that allows a wireless keyboard to communicate with a PC. The standard

originally defined a data rate of 75 kbps for a distance up to 8 m. The Recent standard defines a data rate of 4 Mbps. Infrared signals defined by IrDA transmit through line of sight; the IrDA port on the keyboard needs to point to the PC for transmission to occur.

3.3 Antennas

An antenna can be defined as an electrical conductor or system of conductors used either for radiating electromagnetic energy or for collecting electromagnetic energy.



For transmission of a signal, radio-frequency electrical energy from the transmitter is converted into electromagnetic energy by the antenna and radiated into the surrounding environment (atmosphere, space, water). Reception occurs when the electromagnetic signal intersects the antenna, where the electromagnetic energy is converted into radio-frequency electrical energy and fed into the receiver. In two-way communication, the same antenna can be and often is used for both transmission and reception. This is possible because any antenna transfers energy from the surrounding environment to its input receiver terminals with the same efficiency that it transfers energy from the output transmitter terminals into the surrounding environment, assuming that the same frequency is used in both directions. Put another way, antenna characteristics are essentially the same whether an antenna is sending or receiving electromagnetic energy. An antenna radiates power in all directions but, typically, does not perform equally well in all

directions. A common way to characterize the performance of an antenna is the radiation pattern, which is a graphical representation of the radiation properties of an antenna as a function of space coordinates. The simplest pattern is produced by an idealized antenna known as the **isotropic antenna** radiates power in all directions equally. (Radiation pattern in spherical)

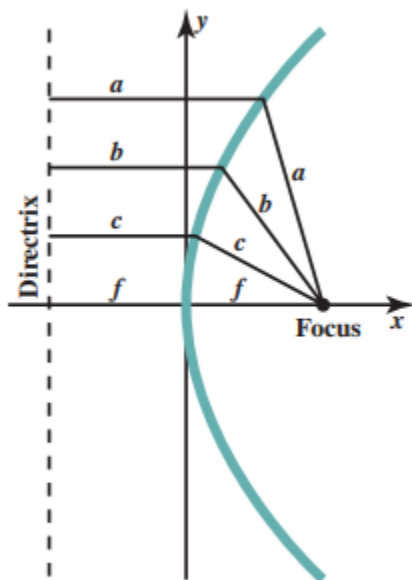
An **omnidirectional antenna**, on the other hand, radiates uniformly in one plane (e.g., horizontally), with the radiation pattern decreasing in the orthogonal (vertical) plane, often having nulls. (Radiation pattern is like a donut).

Parabolic Reflective Antenna

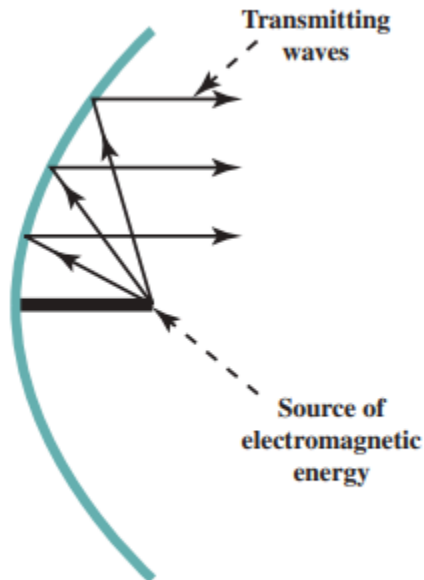
An important type of antenna is the parabolic reflective antenna, which is used in terrestrial microwave and satellite applications. A parabola is the locus of all points equidistant from a fixed line and a fixed point not on the line. The fixed point is called the focus and the fixed line is called the directrix (Figure 4.8a). If a parabola is revolved about its axis, the surface generated is called a paraboloid. A cross section through the paraboloid parallel to its axis forms a parabola and a cross section perpendicular to the axis forms a circle. Such surfaces are used in automobile headlights, optical and radio telescopes, and microwave antennas because of the following property: If a source of electromagnetic energy (or sound) is placed at the focus of the paraboloid, and if the paraboloid is a reflecting surface, then the wave bounces back in lines parallel to the axis of the paraboloid; Figure below shows this effect in cross section. In theory, this effect creates a parallel beam without dispersion. In practice, there is some dispersion, because the source of energy must occupy more than one point. The larger the diameter of the antenna, the more tightly directional is the beam. On reception, if incoming waves are parallel to the axis of the reflecting paraboloid, the resulting signal is concentrated at the focus.

Antenna Gain

Antenna gain is a measure of the directionality of an antenna. Antenna gain is defined as the power output, in a particular direction, compared to that produced in any direction by a perfect omnidirectional antenna (isotropic antenna). Specifically, $G_{dB} = 10 \log (P_2/P_1)$, where G is the antenna gain, P_1 is the radiated power of the directional antenna, and P_2 is the radiated power from the reference antenna. For example, if an antenna has a gain of 3 dB, that antenna improves upon the isotropic antenna in that direction by 3 dB, or a factor of 2. The increased power radiated in a given direction is at the expense of other directions. In effect, increased power is radiated in one direction by reducing the power radiated in other directions. It is important to note that antenna gain does not refer to obtaining more output power than input power but rather to directionality.



(a) Parabola



(b) Cross section of parabolic antenna showing reflective property

Example: Consider a directional antenna that has a gain of 6 dB over a reference antenna and that radiates 700 W. How much power must the reference antenna radiate to provide the same signal power in the preferred direction?

$$\begin{aligned} 6 &= 10 \log(P_2/700) \\ P_2/700 &= 10^{0.6} = 3.98 \\ P_2 &= 2786 \text{ W} \end{aligned}$$

A concept related to that of antenna gain is the effective area of an antenna. The effective area of an antenna is related to the physical size of the antenna and to its shape. The relationship between antenna gain and effective area is:

$$G = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi f^2 A_e}{c^2}$$

where

G = antenna gain (dimensionless number or ratio)

A_e = effective area (m^2)

f = carrier frequency (Hz)

c = speed of light ($\approx 3 \times 10^8 \text{ m/s}$)

λ = carrier wavelength (m)

Expressed in decibels, we have $G_{\text{dB}} = 10 \log G$

Example: For a parabolic reflective antenna with a diameter of 2 m, operating at 12 GHz, what are the effective area and the antenna gain?

area of $A \times \pi r^2 = \pi$ and an effective area of $A_e = 0.56\pi$. The wavelength is $\lambda = c/f = (3 \times 10^8)/(12 \times 10^9) = 0.025\text{m}$. Then

$$G = (7A)/\lambda^2 = (7 \times \pi)/(0.025)^2 = 35,186$$

$$G_{\text{dB}} = 10 \log 35,186 = 45.46 \text{ dB}$$

Free space transmission equation

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi R)^2}$$

Where,

P_r = Power at the receiving antenna

P_t = Output power of transmitting antenna

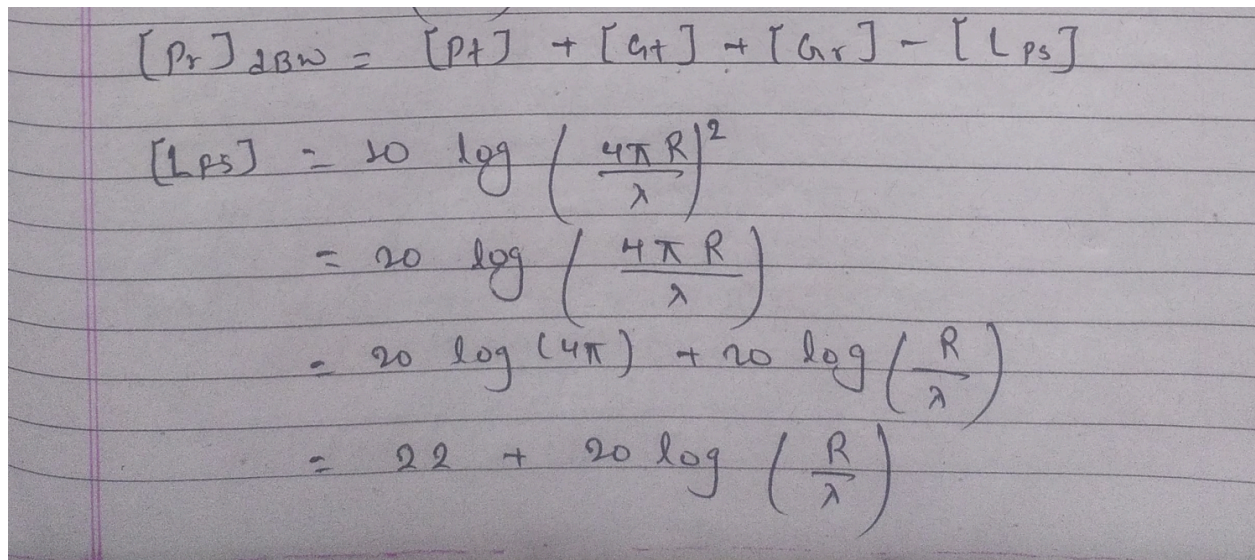
G_t = Gain of the transmitting antenna

G_r = Gain of the receiving antenna

λ = Wavelength

R = Distance between the antennas

The equation can also be represented in dB format:



Handwritten derivation of the Friis transmission equation in dB format:

$$\begin{aligned} [P_r]_{dBW} &= [P_t] + [G_t] + [G_r] - [L_{ps}] \\ [L_{ps}] &= 10 \log \left(\frac{4\pi R}{\lambda} \right)^2 \\ &= 20 \log \left(\frac{4\pi R}{\lambda} \right) \\ &= 20 \log(4\pi) + 20 \log \left(\frac{R}{\lambda} \right) \\ &= 22 + 20 \log \left(\frac{R}{\lambda} \right) \end{aligned}$$

The bracker [] represents dB.

3.3 Satellite communication

->A communication satellite is, in effect, a microwave relay station. It is used to link two or more ground-based microwave transmitter/receivers, known as earth stations, or ground stations.

->The satellite receives transmissions on one frequency band (uplink), amplifies or repeats the signal, and transmits it on another frequency (downlink).

->A single orbiting satellite will operate on a number of frequency bands, called transponder channels, or simply transponders. Figure A depicts in a general way two common configurations for satellite communication. In the first, the satellite is being used to provide a point-to-point link between two distant ground-based antennas. In the second, the satellite provides communications between one ground-based transmitter and a number of ground-based receivers. For a communication satellite to function effectively, it is generally required that it remain stationary with respect to its position over the Earth. Otherwise, it would not be within the line of sight of its earth stations at all times. To remain stationary, the satellite must have a period of rotation equal to the Earth's period of rotation. This match occurs at a height of 35,863 km at the equator. **Two satellites using the same frequency band, if close enough together, interfere with each other. To avoid this, current standards require a 4° spacing (angular displacement as measured from the Earth) in the 4/6-GHz band and a 3° spacing at 12/14 GHz. Thus the number of possible satellites is quite limited.**

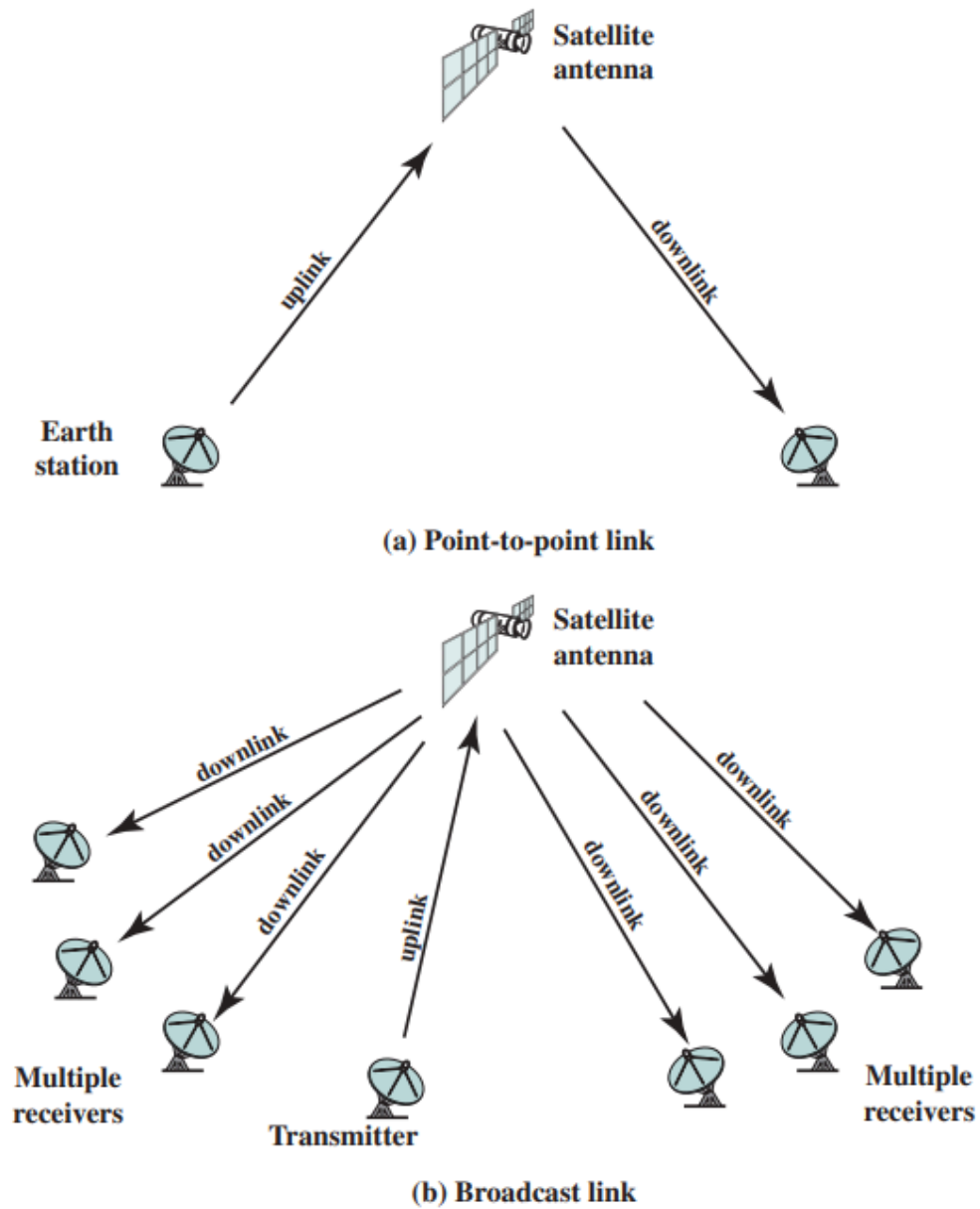


Fig: A

Applications:

- Television distribution
- Long-distance telephone transmission
- Private business networks
- Global positioning

Because of their broadcast nature, satellites are well suited to television distribution and are being used extensively in the United States and throughout the world for this purpose. In its traditional use, a network provides programming from a central location. Programs are transmitted to the satellite and then broadcast down to a number of stations, which then distribute the programs to individual viewers. One network, the Public Broadcasting Service (PBS), distributes its television programming almost exclusively by the use of satellite channels. Other commercial networks also make substantial use of satellites, with cable television systems receiving an ever-increasing proportion of their programming from satellites. The most recent application of satellite technology to television distribution is direct broadcast satellite (DBS), in which satellite video signals are transmitted directly to the home user. The decreasing cost and size of receiving antennas have made DBS economically feasible.

Satellite transmission is also used for point-to-point trunks between telephone exchange offices in public telephone networks. It is the optimum medium for high-usage international trunks and is competitive with terrestrial systems for many long-distance intranational links. There are a number of business data applications for satellites. The satellite provider can divide the total capacity into a number of channels and lease these channels to individual business users. A user equipped with the antennas at a number of sites can use a satellite channel for a private network. Traditionally, such applications have been quite expensive and limited to larger organizations with high-volume requirements. A recent development is the very small aperture terminal (VSAT) system, which provides a low-cost alternative. Figure B depicts a typical VSAT configuration. A number of subscriber stations are equipped with low-cost VSAT antennas. Using some discipline, these stations

share a satellite transmission capacity for transmission to a hub station. The hub station can exchange messages with each of the subscribers and can relay messages between subscribers.

A final application of satellites, which has become pervasive, is worthy of note. The Navstar Global Positioning System, or GPS for short, consists of three segments or components:

- A constellation of satellites (currently 27) orbiting about 20,000 km above the Earth's surface, which transmits ranging signals on two frequencies in the microwave part of the radio spectrum
- A control segment which maintains GPS through a system of ground monitor stations and satellite upload facilities
- The user receivers—both civil and military

Each satellite transmits a unique digital code sequence of 1s and 0s, precisely timed by an atomic clock, which is picked up by a GPS receiver's antenna and matched with the same code sequence generated inside the receiver. By lining up or matching the signals, the receiver determines how long it takes the signals to travel from the satellite to the receiver. These timing measurements are converted to distances using the speed of light. Measuring distances to four or more satellites simultaneously and knowing the exact locations of the satellites (included in the signals transmitted by the satellites), the receiver can determine its latitude, longitude, and height while also synchronizing its clock with the GPS time standard which also makes the receiver a precise timepiece.

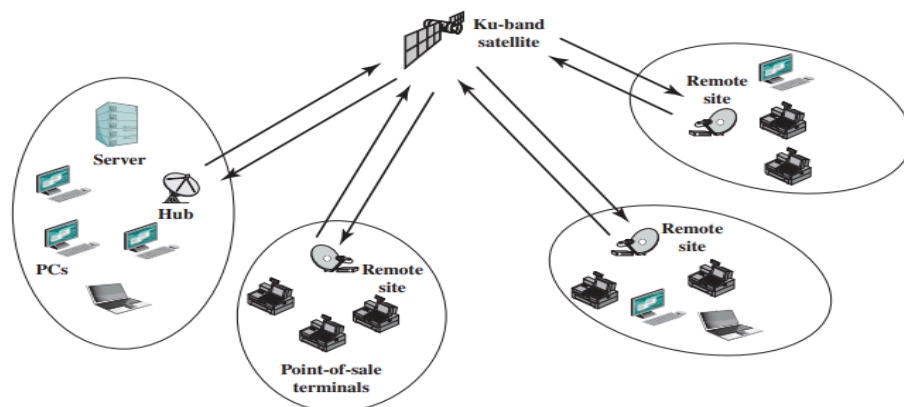


Fig: B

Transmission Characteristics:

The optimum frequency range for satellite transmission is in the range 1 to 10 GHz. Below 1 GHz, there is significant noise from natural sources, including galactic, solar, and atmospheric noise, and human-made interference from various electronic devices. Above 10 GHz, the signal is severely attenuated by atmospheric absorption and precipitation.

->Most satellites providing point-to-point service today use a frequency bandwidth in the range 5.925 to 6.425 GHz for transmission from Earth to satellite (uplink) and a bandwidth in the range 3.7 to 4.2 GHz for transmission from satellite to Earth (downlink). This combination is referred to as the 4/6-GHz band.

->Note that the uplink and downlink frequencies differ. For continuous operation without interference, a satellite cannot transmit and receive on the same frequency.

->Thus signals received from a ground station on one frequency must be transmitted back on another. The 4/6 GHz band is within the optimum zone of 1 to 10 GHz but has become saturated. Other frequencies in that range are unavailable because of sources of interference operating at those frequencies, usually terrestrial microwave. Therefore, the 12/14-GHz band has been developed (uplink: 14 to 14.5 GHz; downlink: 11.7 to 12.2 GHz). At this frequency band, attenuation problems must be overcome. However, smaller and cheaper earth-station receivers can be used. It is anticipated that this band will also saturate, and use is projected for the 20/30-GHz band (uplink: 27.5 to 30.0 GHz; downlink: 17.7 to 20.2 GHz). This band experiences even greater attenuation problems but will allow greater bandwidth (2500 MHz versus 500 MHz) and even smaller and cheaper receivers. Several properties of satellite communication should be noted. First, because of the long distances involved, there is a propagation delay of about a quarter second from transmission from one earth station to reception by another earth station. This delay is noticeable in ordinary telephone conversations. It also introduces problems in the areas of error control and flow control, which we discuss in later chapters. Second, a satellite microwave is inherently a broadcast facility. Many stations can

transmit to the satellite, and a transmission from a satellite can be received by many stations.

Type of satellite:

Types based on orbit:

- **Low Earth Orbit (LEO):** Satellites in LEO orbit at low altitudes, typically between 160 and 2,000 kilometers. LEO satellites are often used for communication, navigation, and Earth observation.
- **Medium Earth Orbit (MEO):** Satellites in MEO orbit at altitudes between 2,000 and 35,786 kilometers. MEO satellites are commonly used for navigation systems like GPS.
- **Geostationary Orbit (GEO):** Satellites in GEO orbit at an altitude of approximately 35,786 kilometers, maintaining a constant position relative to the Earth's surface. GEO satellites are frequently used for communication and weather monitoring

3.3 Bluetooth

Bluetooth is an always-on, short-range radio hookup that resides on a microchip. The concept behind Bluetooth is to provide a universal short-range wireless capability. Using the 2.4-GHz band, available globally for unlicensed low-power

uses, two Bluetooth devices within 10 m of each other can share up to 720 kbps of capacity. Bluetooth is intended to support an open-ended list of applications, including data (e.g., schedules and telephone numbers), audio, graphics, and even video. For example, audio devices can include headsets, cordless and standard phones, home stereos, and digital MP3 players. Examples of some of the capability Bluetooth can provide consumers are as follows:

- Make calls from a wireless headset connected remotely to a cell phone.
- Eliminate cables linking computers to printers, keyboards, and the mouse.
- Hook up MP3 players wirelessly to other machines to download music.
- Set up home networks so that a couch potato can remotely monitor air conditioning, the oven, and children's Internet surfing.
- Call home from a remote location to turn appliances on and off, set the alarm, and monitor activity.

Bluetooth Radio Specification

The Bluetooth radio specification is a short document that gives the basic details of radio transmission for Bluetooth devices. Some of the key parameters are summarized in Table 18.4. Bluetooth makes use of the 2.4-GHz band within the ISM (Industrial, Scientific, and Medical) band. In most countries, the bandwidth is sufficient to define 79 1-MHz physical channels. Power control is used to keep the devices from emitting any more RF power than necessary. The power control algorithm is implemented using the link management protocol between a master and the slaves in a piconet. Modulation for Bluetooth is Gaussian FSK, with a binary one represented by a positive frequency deviation and a binary zero represented by a negative frequency deviation from the center frequency. The minimum deviation is 115 kHz.

Bluetooth Baseband Specification

One of the most complex of the Bluetooth documents is the baseband specification. In this section we provide an overview of the key elements.

Frequency Hopping

Frequency hopping (FH) in Bluetooth serves two purposes:

1. It provides resistance to interference and multipath effects.
2. It provides a form of multiple access among co-located devices in different piconets.

The FH scheme works as follows. The total bandwidth is divided into 79 (in almost all countries) physical channels, each of bandwidth 1 MHz. FH occurs by jumping from one physical channel to another in a pseudorandom sequence. The same hopping sequence is shared by all of the devices on a single piconet; we will

Bluetooth Radio and Baseband Parameters

Topology	Up to 7 simultaneous links in a logical star
Modulation	GFSK
Peak data rate	1 Mbps
RF bandwidth	220 kHz (−3 dB), 1 MHz (−20 dB)
RF band	2.4 GHz, ISM band
RF carriers	23/79
Carrier spacing	1 MHz
Transmit power	0.1 W
Piconet access	FH-TDD-TDMA
Frequency hop rate	1600 hops/s
Scatternet access	FH-CDMA

refer to this as an FH channel. 1 The hop rate is 1600 hops per second, so that each physical channel is occupied for a duration of 0.625 ms. Each 0.625-ms time period is referred to as a slot, and these are numbered sequentially. Bluetooth radios communicate using a time division duplex (TDD) discipline. Because more than two devices share the piconet medium, the access technique is TDMA. Thus piconet access can be characterized as FH-TDD-TDMA.

3.3 WIFI

WiFi is a wireless technology that allows electronic devices to connect to the internet and communicate with each other without a physical cable. This uses radio waves to transmit the data between a WiFi router and compatible devices like

smartphones, computers, and smart home gadgets. These WiFi networks are common in homes, offices, and public spaces providing convenient internet access and local connectivity. This technology has become an essential part of modern digital life enabling wireless internet browsing, file sharing, and device communication in various settings.

The Wireless Access Point (WAP) means a wireless access point is the networking device that allows WiFi-enabled devices to connect to the wired network. It acts as the central hub broadcasting the Wi-Fi signal that devices can detect and join. Access points are commonly used to extend the range of existing networks, create separate network segments, or provide Wi-Fi connectivity in large spaces like offices, schools or public areas. They can be standalone devices or integrated into a router and they enable multiple devices to access network resources and the internet wirelessly within their coverage area.

Working:

Wi-Fi is a wireless technology for networking, so it uses Electromagnetic waves to transmit networks. We know that there are many divisions of Electromagnetic waves according to their frequency such as X-ray, Gamma-ray, radio wave, microwave, etc, in Wi-Fi, the radio frequency is used. For transmitting Wi-Fi signal there is three medium,

- **Base Station Network or an Ethernet(802.3) Connection:** It is the main host network from where the network connection is provided to the router.
- **Access Point or Router:** it is a bridge between a wired network and a wireless network. It accepts a wired Ethernet connection and converts the wired connection to a wireless connection and spreads the connection as a radio wave.

- **Accessing Devices:** It is our mobile, computer, etc from where we use the Wi-Fi and surfing internet.

All the electronics devices read data in binary form, also the router or our devices, here routers provide radio waves and those waves are received by our devices and read the waves in binary form. We know how a wave looks like a upper pick of the wave is known as a 1 and the lower pick of the wave is known as the 0 in a binary as mentioned below :

- **SSID (Service Set Identifier):** This 32 character name that identifies the Wi-Fi network and differentiates the one Wi-Fi from another Wi-Fi. All the devices are attempting to connect a particular SSID. A SSID is the name of a wireless network.
- **WPA-PSK (Wi-Fi Protected Access- Pre-Shared Key):** The program developed by Wi-Fi Alliance Authority to secure wireless networks with the use of a Pre-Shared Key authentication. The WPA has 3 types such as WPA, WPA2 and a WPA3. It is a way of encrypting a Wi-Fi signal to protect from unwanted users.
- Wi-Fi uses Ad-Hoc networks to transmit. It is a point-to-point network without any interface.

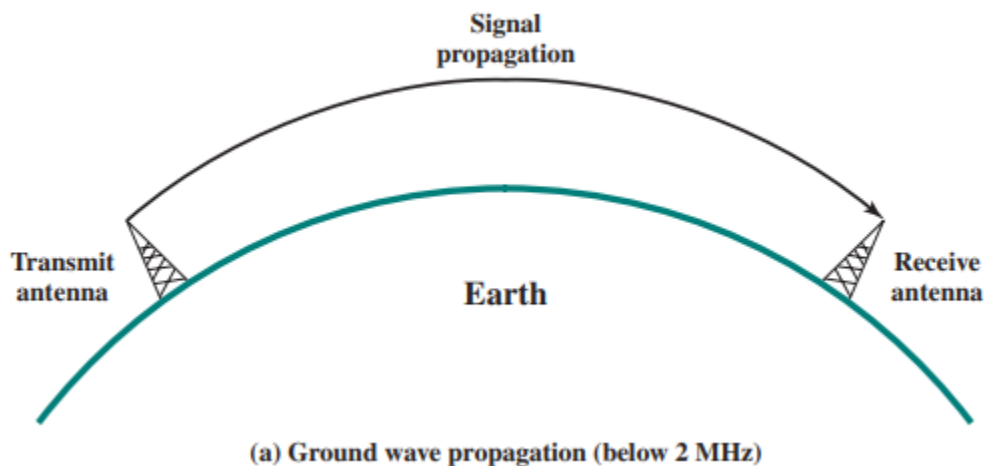
3.4 Wireless Propagation

A signal radiated from an antenna travels along one of three routes: ground wave, sky wave, or line of sight (LOS). Table 4.5 shows in which frequency range each predominates. In this book, we are almost exclusively concerned with LOS communication, but a short overview of each mode is given in this section.

Ground Wave Propagation

Ground wave propagation more or less follows the contour of the Earth and can propagate considerable distances, well over the visual horizon. This effect is found in frequencies up to about 2 MHz. Several factors account for the tendency of electromagnetic waves in this frequency band to follow the Earth's curvature. One factor is that the electromagnetic wave induces a current in the Earth's surface, the result of which is to slow the wavefront near the Earth, causing the wavefront to tilt downward and hence follow the Earth's curvature. Another factor is diffraction, which is a phenomenon having to do with the behavior of electromagnetic waves in the presence of obstacles. Electromagnetic waves in this frequency range are scattered by the atmosphere in such a way that they do not penetrate the upper atmosphere.

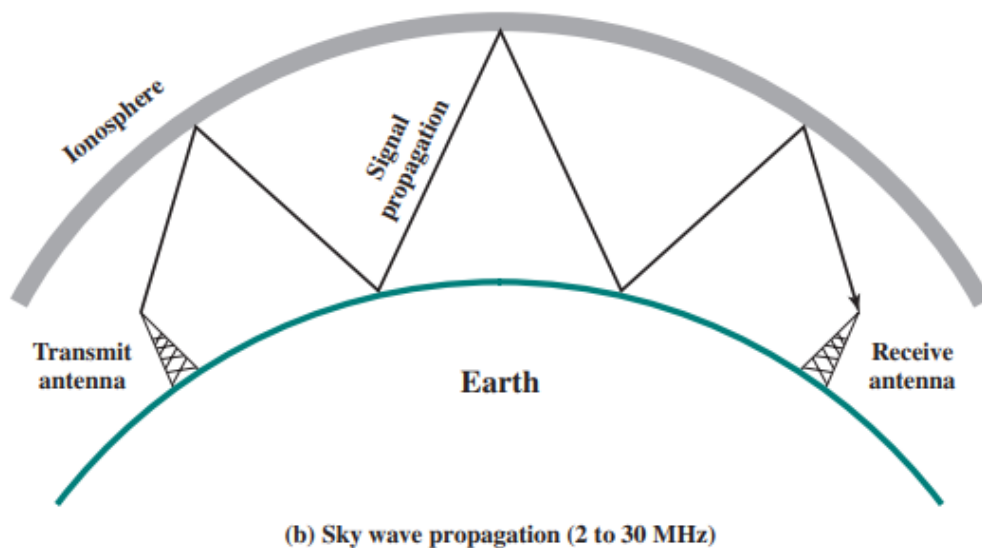
The best-known example of ground wave communication is AM radio.



Sky Wave Propagation:

Frequ range: 2 Mhz - 30 Mhz

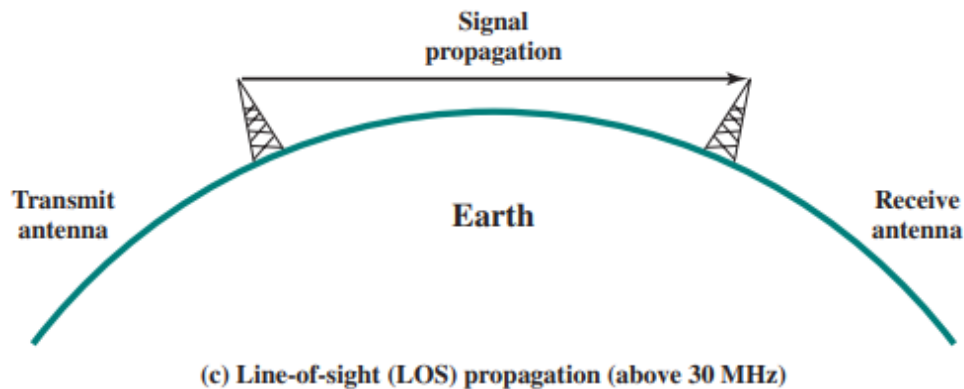
Sky wave propagation is used for amateur radio and international broadcasts such as BBC and Voice of America. With sky wave propagation, a signal from an earth-based antenna is reflected from the ionized layer of the upper atmosphere (ionosphere) back down to Earth. Although it appears the wave is reflected from the ionosphere as if the ionosphere were a hard reflecting surface, the effect is in fact caused by refraction. Refraction is described subsequently. A sky wave signal can travel through a number of hops, bouncing back and forth between the ionosphere and the Earth's surface. With this propagation mode, a signal can be picked up thousands of kilometers from the transmitter.



Line-of-Sight Propagation:

Above 30 MHz, neither ground wave nor sky wave propagation modes operate, and communication must be by line of sight. For satellite communication, a signal above 30 MHz is not reflected by the ionosphere and therefore a signal can be

transmitted between an earth station and a satellite overhead that is not beyond the horizon. For ground-based communication, the transmitting and receiving antennas must be within an effective line of sight of each other. The term effective is used because microwaves are bent or refracted by the atmosphere. The amount and even the direction of the bend depend on conditions, but generally microwaves are bent with the curvature of the Earth and will therefore propagate farther than the optical line of sight.



3.4 Frequency Band

<i>Band</i>	<i>Range</i>	<i>Propagation</i>	<i>Application</i>
VLF (very low frequency)	3-30 kHz	Ground	Long-range radio navigation
LF (low frequency)	30-300 kHz	Ground	Radio beacons and navigational locators
MF (middle frequency)	300 kHz-3 MHz	Sky	AM radio
HF (high frequency)	3-30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30-300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz-3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3-30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30-300 GHz	Line-of-sight	Radar, satellite

3.5 Error detection and correction

a. Parity

- It can detect single bit error
- Simple but poor protection against transmission error
- Only suitable for short and less error prone transfer path
- **There are 5 types of parity**
- No parity: No parity is used

- Even Parity: Numbers of 1's is made even including parity bit
- Odd Parity: Numbers of 1's is made odd including parity bit
- Mark parity: Parity bit is always set to 1
- Space parity: Parity bit is always set to 0

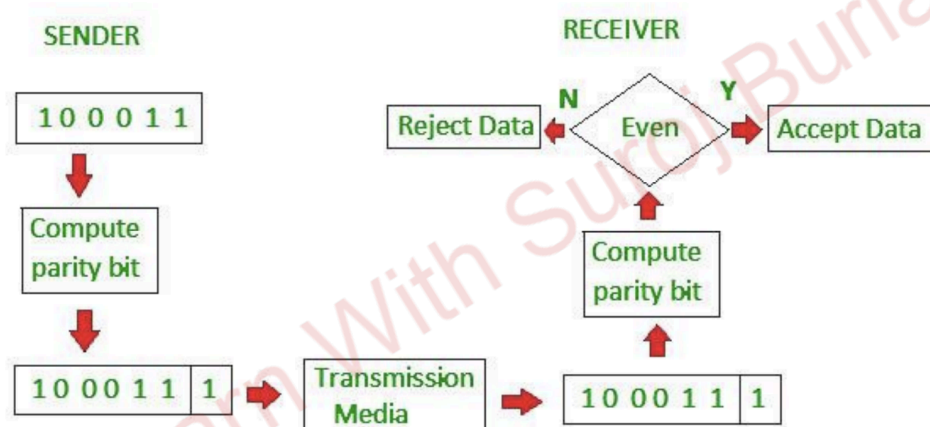
Note: Parity can detect only odd number of errors. It cannot detect even number of errors.

For example : Suppose we are using even parity. The original message is 1011. Hence using even parity, the total 1's should be even. Hence the data to be transmitted is 10111. Where the one at the LSB is parity bit.

If error occurs and the third bit is flipped , the received data becomes 10011. That is the 1's in the received data becomes odd. That is the receiver will know error has occurred as even parity was used.

But if bit 2 is also flipped along with bit 3 during transmission, the received data becomes 10001. Here despite of 2 errors, the total number of 1's are even . So, the receiver will not be able to detect the error.

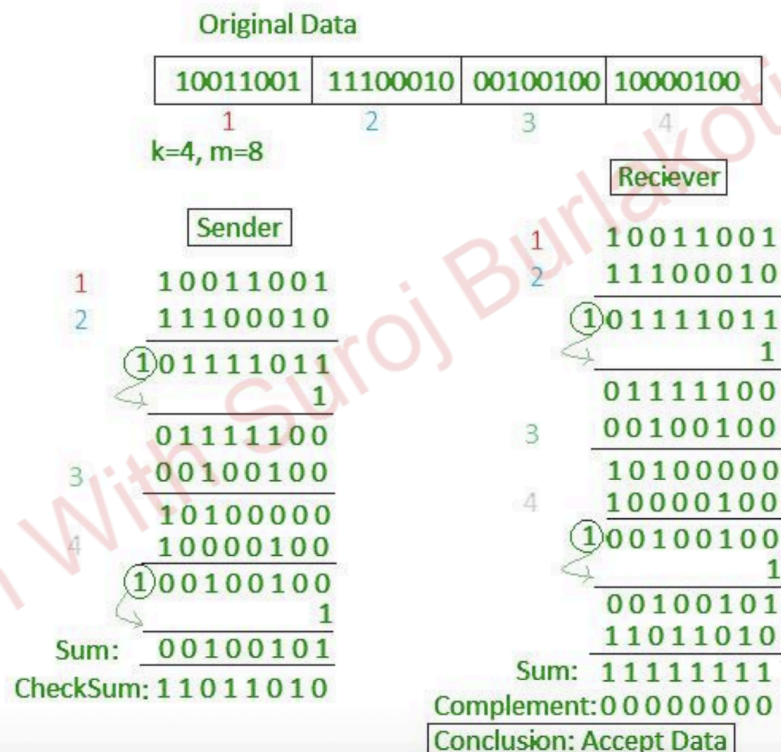
Parity Check



Checksum

- The checksum technique is used when blocks of data are transmitted.
- At sender side,
 - If m bit checksum is used, the data unit to be transmitted is divided into segments of m bits.
 - All the m bit segments are added.
 - The result of the sum is then complemented using 1's complement arithmetic.
 - The value so obtained is called as **checksum**.
- The data along with the checksum value is transmitted to the receiver.
- At receiver side,
 - If m bit checksum is being used, the received data unit is divided into segments of m bits.
 - All the m bit segments are added along with the checksum value.
 - The value so obtained is complemented and the result is checked.
- If the result is zero,
 - Receiver assumes that no error occurred in the data during the transmission.
 - Receiver accepts the data.
- If the result is non-zero
 - Receiver assumes that error occurred in the data during the transmission.
 - Receiver discards the data and asks the sender for retransmission.

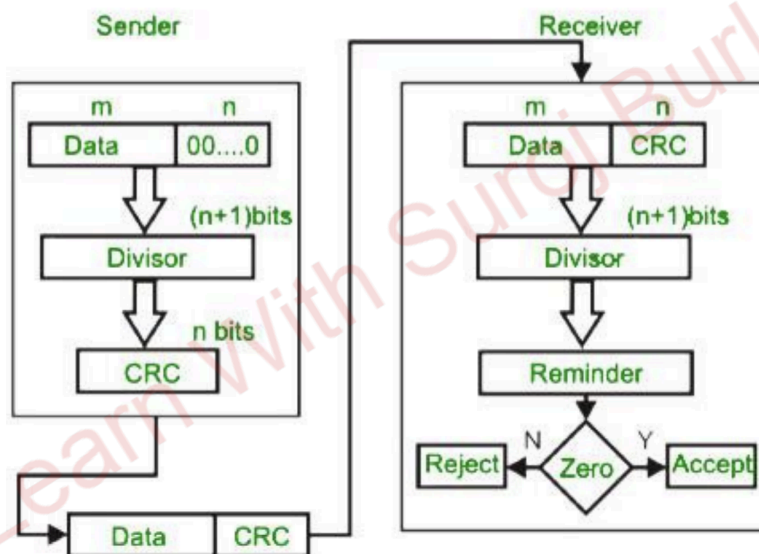
Checksum



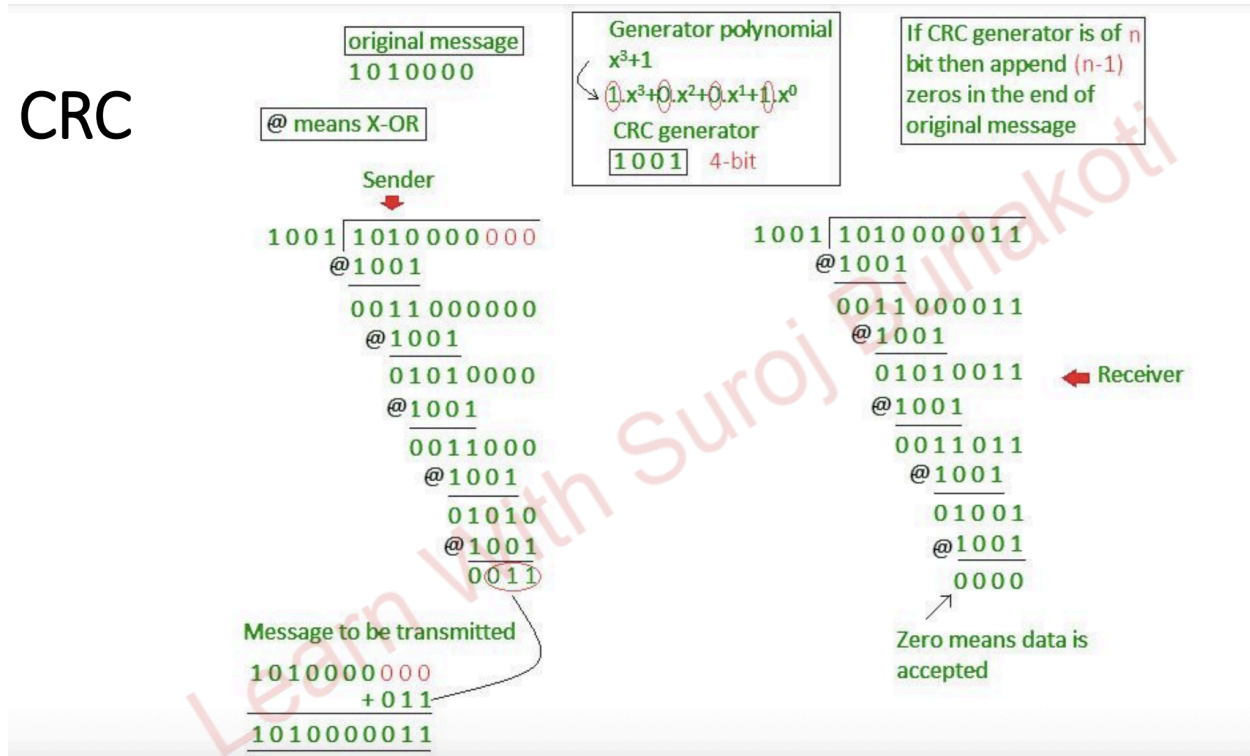
Cyclic Redundancy Check (CRC)

- This technique is based on mathematical relationships of polynomials.
- A stream of data can be represented as a polynomial that is divided by a constant polynomial, and the remainder, unique to that set of bits, is generated.
- The remainder is sent out as a check for errors.
- The receiver checks the remainder to detect an error in the transmission.
- This is a somewhat complex technique for error checking.

Cyclic Redundancy Check (CRC)



CRC



Hamming code

Reference: [▶ Hamming Code | Error Detection](#)

[▶ Hamming Code | Error Correction](#)

- Hamming Code is an error detection and correction method.
- Invented by Richard W. Hamming in 1950.
- It adds extra parity bits to data bits to enable error correction.
- It can detect and correct single-bit errors.
- It can detect (but not correct) two-bit errors.
- Parity bits are placed at bit positions that are powers of 2 (1, 2, 4, 8, etc.).
- Each parity bit checks specific positions in the data to identify errors.
- The most common form is the (7,4) Hamming code, which has 4 data bits and 3 parity bits.

- Errors are located using a binary syndrome generated during checking.
- Hamming codes are widely used in memory, digital communication, and data transmission systems.

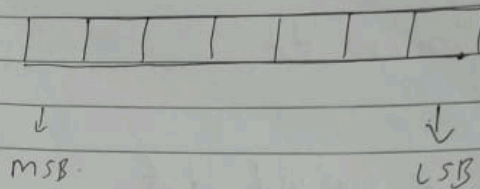
Contd.....

Method	Error Detection Capability	Error Correction Capability
Parity Bit	Detects single-bit errors only	None
Checksum	Detects many single-bit and some multi-bit errors (not guaranteed)	None
CRC	Detects burst errors (up to length $n-k$, depends on polynomial)	None (correction possible in special cases with extra logic)
Hamming Code (7,4)	Detects up to 2-bit errors	Corrects 1-bit errors

Hamming code example Numerical :

Given by P.W hamming
 Hamming code $\rightarrow$ Easy to implement
 $\rightarrow$ 7-bit hamming code is commonly used i.e. 4 bit-data 3 bit parity.

① Error detection



$\Rightarrow$ Position of parity bit is determined by

$$2^n = \{ \text{where } n = 0, 1, \dots, n \}$$

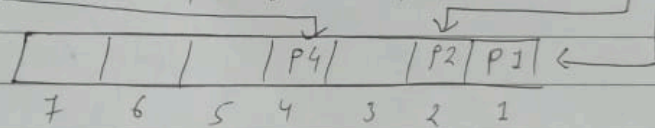
i.e. $2^0 = 1$

$2^1 = 2$

$2^2 = 4$

$2^3 = 8$

Hence parity bit position are



No 8th position is available as Total bit = 7

$\Rightarrow$ other bits are data i.e. D_3, D_5, D_6, D_7 are data bits whose value determine the parity bit's value.

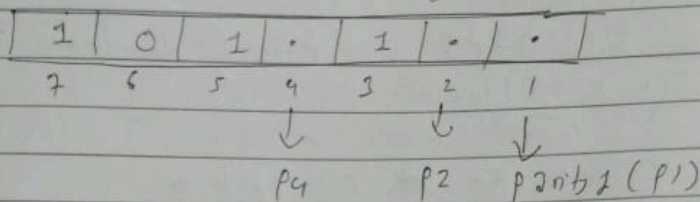
$$P1 \leftarrow D_3 D_5 D_7$$

$$P2 \leftarrow D_3 D_6 D_7$$

$$P4 \leftarrow D_5 D_6 D_7$$

Eg: 1011 is the data to be transmitted

So, let's construct the hamming code



Now, for even parity

$P_1 = (111)$ must be 1

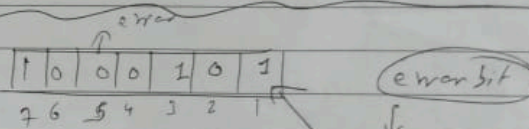
$P_2 = (101)$ must be 0

$P_4 = (101)$ must be ~~zero~~ 0

So

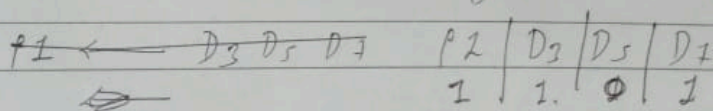


Error correction



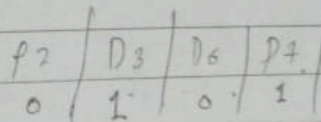
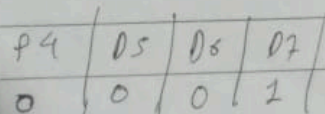
Suppose the received data is = 1001

then, let's see how hamming code corrects it



Contradiction $\Rightarrow$ set parity = 1
 No contradiction $\Rightarrow$ parity = 0

$\Rightarrow$ parity is ~~even~~ odd
 $\Rightarrow$ no contradiction
 $\Rightarrow$ Hence put $P_1 = 1$



$\Rightarrow$ odd parity
 $\Rightarrow$ contradiction
 $\Rightarrow$ put $P_4 = 1$

$\Rightarrow$ Even parity
 $\Rightarrow$ no contradiction
 $\Rightarrow$ put $P_2 = 0$

Hence

$$\begin{aligned} & (149291) \\ & = (101)_{10} \quad \text{Binary to decimal} \\ & = \underline{\underline{(5)_{10}}} \end{aligned}$$

∴ error is at ~~1st~~ 5th position

∴ Hence, the correction is to flip the bit such that the corrected data is

$$\Rightarrow \boxed{1010101}$$

3.6 Data compression [Lossy and Lossless]

What is Data Compression?

Data compression is the process of reducing the size of data so that it takes up less storage space or requires less bandwidth for transmission. It works by removing redundancies or unnecessary information from the data.

Why is Compression Required?

- To save **storage space**
- To reduce **transmission time**
- To lower **bandwidth usage**
- Essential for **real-time applications** like video conferencing, streaming, etc.

Example: Raw Video Size Calculation

Let's calculate the size of a **raw, uncompressed video**:

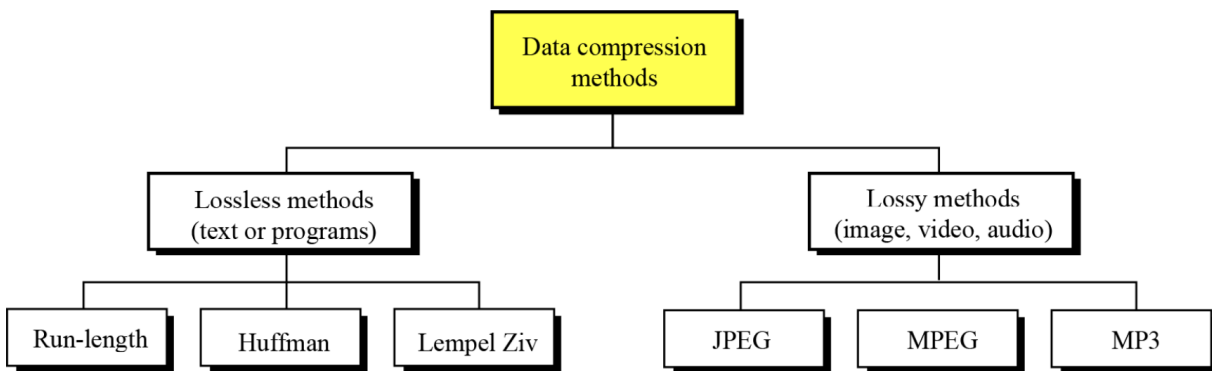
- Resolution: 1920×1080 (Full HD)
- Frame rate: 30 frames/sec
- Color depth: 24 bits/pixel (8 bits for R, G, B)

Size per frame = $1920 \times 1080 \times 24$ bits
= 49,766,400 bits ≈ 6.2 MB
Per second = $6.2 \text{ MB} \times 30 \approx 186 \text{ MB/sec}$

For 1 minute = $186 \times 60 = 11.2 \text{ GB}$

So, streaming raw video would need huge bandwidth (~1.5 Gbps), which is not practical for real-time transmission. Hence, compression is essential.

Types of Compression



1. Lossless Compression

- No information is lost.
- The original data can be perfectly reconstructed.
- Used where **accuracy is important** (e.g., text files, medical images).
- Examples: ZIP, Run-length, Huffman

2. Lossy Compression

- Some data is **permanently removed**, especially parts not easily perceived.
- The original data **cannot be fully recovered**.
- Used in **audio, image, and video** (e.g., MP3, JPEG, MPEG).

- Achieves **higher compression ratios** than lossless.

Lossless compression

1. Run Length Encoding (RLE)

- RLE is a simple **lossless data compression** technique.
- It replaces **consecutive repeated values** (runs) with a **single value and the number of repetitions**.
- Works best for data with **long runs of the same value**, such as simple images or binary data.

How RLE Works

- Instead of storing all repeated values, store:
 - The **value**.
 - The **count** of how many times it repeats.

Example

Original data:

A A A A B B C C C C C

RLE encoded:

(A, 4), (B, 2), (C, 5)

Another example with binary image data:

Original: 1 1 1 0 0 1 1 0 0 0 0

Encoded: (1,3), (0,2), (1,2), (0,4)

Advantages

- Easy to implement.
- Effective for data with lots of repetition (e.g., black-and-white images, icons, simple graphics).

Disadvantages

- Inefficient for data without repetition.
- May increase file size if the data has frequent value changes.

2. Huffman Encoding (Entropy Encoding)

Huffman encoding is a widely used lossless data compression algorithm that assigns variable-length codes to input characters, with shorter codes assigned to more frequent characters and longer codes to less frequent ones. This technique reduces the overall number of bits required to represent the data. The process begins by analyzing the frequency of each character in the data set, then constructing a binary tree called a Huffman tree. Each leaf node of the tree represents a character, and the path from the root to the leaf determines the binary code for that character. Huffman encoding is optimal when character frequencies are known and is commonly used in formats like ZIP files and JPEG image compression.

⑧ Huffman coding

1. List symbols in decreasing order of probability.
2. Combine two ~~at~~ least probable symbol & rearrange.
3. Continue reduction until the no. of reduced symbol is 2.
4. Assign bit zero(0) to highest probability and one(1) to lowest.
5. Go back to combined symbols and add second bit as '0' & '1' & retaining all assignments made before.
6. keep regressing this way until the first column is reached.

Example:

x_i	$P(x_i)$							(n) length
x_1	0.30	00	00	00	0.30	0.45	0.55	2
x_2	0.25	01	01	01	0.25	0.30	0.45	2
x_3	0.20	11	11	11	0.25	0.25		2
x_4	0.12	101	101	100	0.20	0.20		3
x_5	0.08	1000	1000	101	0.12			4
x_6	0.05	1001	1001					4

Formulas for huffman coding

Now:

$$\text{Entropy } H(x) = \sum_{i=1}^n P(x_i) \log_2 \left(\frac{1}{P(x_i)} \right)$$

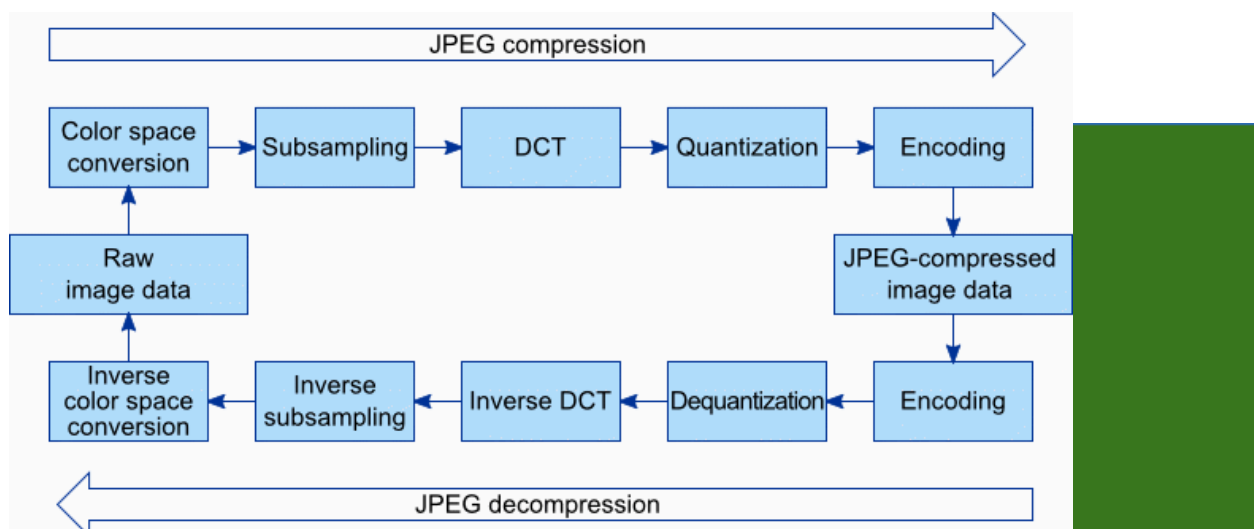
$$\text{Average length } (L) = \sum_{i=1}^n P(x_i) n_i$$

$$\text{code efficiency } (\eta) = \frac{H(x)}{L}$$

Lossy Compression

1. Image compression (JPEG)

Reference video: [📺 How Image Compression Works](#)



JPEG Compression Process – Explained in Bullet Points

1. Color Space Conversion:

Converts image from RGB to YCbCr.

Y = Luminance (brightness), Cb and Cr = Chrominance (color info).

Human eyes are more sensitive to brightness than color, so this separation is important.

2. Chroma Subsampling:

Reduces resolution of Cb and Cr components

This means color info is stored at a lower resolution than brightness.

Saves space with minimal visual quality loss.

3. Block Splitting:

The image is divided into 8×8 pixel blocks.

Compression is applied separately to each block.

4. Discrete Cosine Transform (DCT):

Converts the spatial pixel values into frequency components.

The result is a block with:

Low-frequency components – important for image structure.

High-frequency components– represent fine details/noise.

5. Quantization:

->More aggressive compression on high-frequency components.

->Since human eyes can't see fine details well, these values are divided by a quantization matrix and often become zero.

->This step introduces loss, making JPEG a lossy compression.

6. Zigzag Scanning:

Rearranges the quantized block into a 1D array.

Groups zeros together to improve compression.

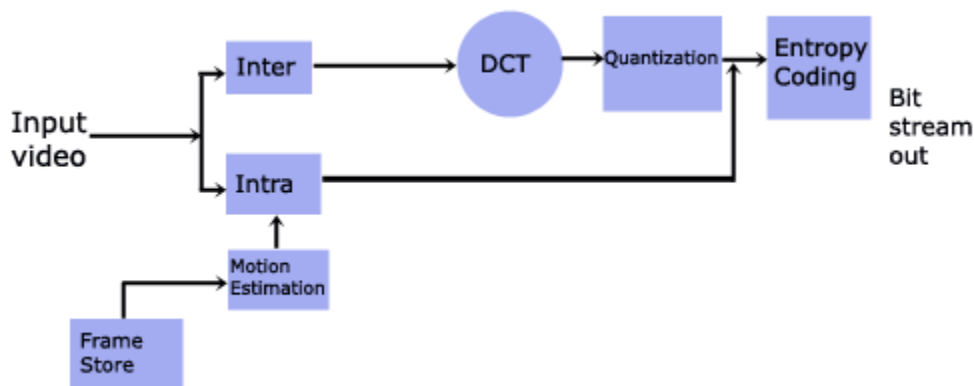
7. Use run-length encoding

8. Entropy Coding (e.g., Huffman Coding):

Final step: compresses the data using lossless encoding.

Takes advantage of patterns and repeated values (like zeros).

2. Video compression (MPEG)



Aspect	Intra-frame (I-frames)	Inter-frame (P-frames & B-frames)
--------	------------------------	-----------------------------------

Definition	Compression within a single frame	Compression using differences between frames
Also called	Spatial compression	Temporal compression
Reference	Self-contained (does not need other frames)	Depends on one or more other frames
Compression Ratio	Lower (less compressed)	Higher (more compressed)
Usage	Used as reference points (keyframes)	Used to save data between keyframes
File Size	Larger	Smaller
Processing Time	Faster decoding	Slower decoding (needs reference frame)

How MPEG Compression Works

1. Each Frame is an Image

- MPEG compresses **video** as a sequence of **individual frames (images)**.
- Each frame can be compressed using **JPEG-like** techniques (especially **intra-coded I-frames**).

2. Spatial Redundancy Removal (Within a Frame)

- Each **I-frame** is compressed **independently**, just like a JPEG image:
 - Converts to **YUV** color space.
 - Applies **chroma subsampling** (since human eyes are less sensitive to color).

- Uses **DCT (Discrete Cosine Transform)** to separate visible and invisible frequencies.
- **Quantizes** and **encodes** the DCT coefficients.

3. Temporal Redundancy Removal (Between Frames)

- **Most consecutive frames are similar** (like in slow scenes or static backgrounds).
- Instead of storing every full frame, MPEG uses:
 - **P-frames (Predicted frames):**
 - Store **only the difference** (called **residual**) from a previous I- or P-frame.
 - If part of the frame is the same, it's not stored again.
 - Saves space.
 - **B-frames (Bidirectional predicted frames):**
 - Predict from **both previous and next frames**.
 - More compression, but require more processing.

4. Run-Length Encoding (for Repeated Frames or Blocks)

- If multiple consecutive frames or blocks are **identical**:
 - MPEG can store them as:
Repeat (frame/block, x times)
(like **Run-Length Encoding**).
 - Common in static scenes or freeze frames.

♦ 5. Motion Compensation

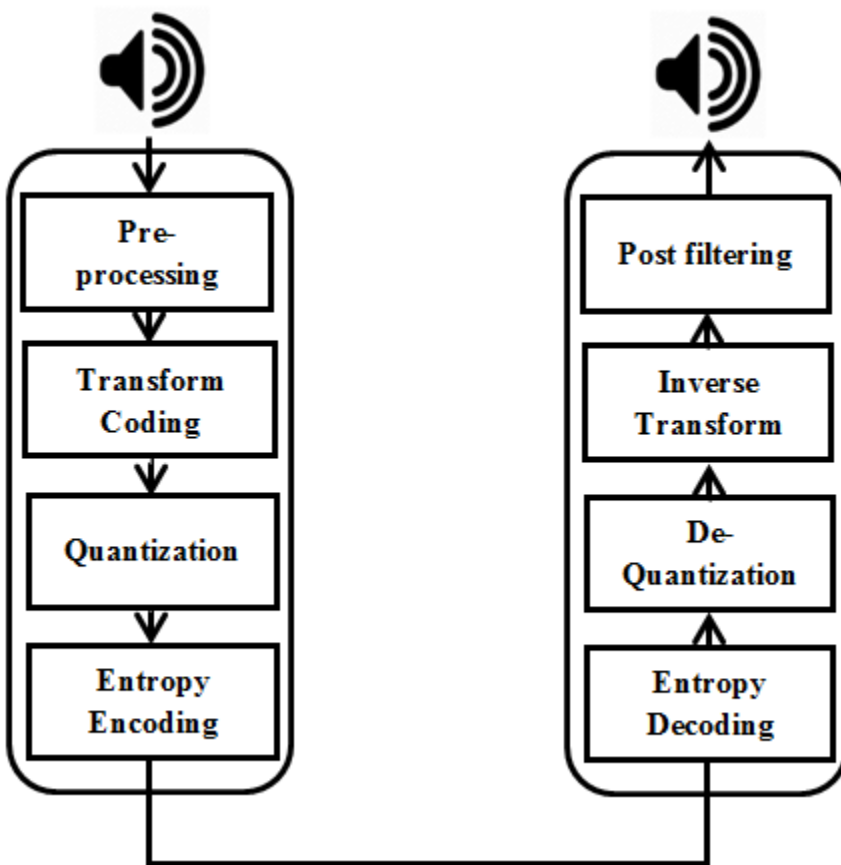
- For moving objects, MPEG finds **motion vectors**:
 - Describes how parts of an image move from one frame to the next.

- Instead of storing new pixel data, MPEG stores **how the object moved**.
- The difference is encoded in a **residual frame**.

3. Audio compression (mp3)

Reference video:

📺 [How MP3 File Works | MP3 Compression Explained In 3 Minutes](#)



Steps in MP3 Compression:

1. Segmentation of Audio Signal

- The raw audio signal is divided into small frames (usually 576 or 1152 samples per frame).

2. Frequency Transformation

- Each frame is transformed from the **time domain to the frequency domain** using the **Modified Discrete Cosine Transform (MDCT)**.
- This helps identify the frequency components present in the signal.

3. Psychoacoustic Modeling

- MP3 uses psychoacoustic models to determine which sounds are **audible** to the human ear.
- Sounds that are **masked** by louder nearby frequencies are either removed or heavily compressed.
- This step eliminates parts of the audio that are unlikely to be heard.

4. Quantization

- The remaining frequency components are quantized (converted into a finite set of values), reducing precision and data size.

5. Huffman Coding

- The quantized data is further compressed using **Huffman coding**, a lossless algorithm that assigns shorter codes to more frequent patterns.

6. Bitstream Formatting

- The compressed data is packaged into frames, including side information like bit allocation and scale factors, to reconstruct the signal later.

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Smample question from old question.

6. Compare the transmission characteristics and performance (frequency range, bandwidth, security, flexibility, interference, connectivity) of Optical **fiber** cable and Satellite transmission. [6]

Parameter	Optical Fiber Cable	Satellite Transmission
Frequency Range	Operates in the range of THz (Infrared and visible light)	Uses GHz range (typically 1–40 GHz)
Bandwidth	Very high bandwidth , supports multiple Tbps (Terabits/sec)	Limited bandwidth , generally a few hundred Mbps to a few Gbps
Security	Highly secure , difficult to tap without being detected	Less secure , signals can be intercepted with special equipment
Flexibility	Less flexible in deployment; physical installation required	Highly flexible , covers remote and mobile areas without wiring
Interference	Low interference , immune to electromagnetic interference	High susceptibility to weather (rain fade), atmospheric conditions
Connectivity	Point-to-point , suitable for local and long-haul connections	Wide coverage area , supports broadcast/multicast to many users
Latency	Low latency , especially over short distances	High latency , typically ~250–600 ms due to signal travel time

Installation Cost	High initial cost , but low maintenance	High setup cost (satellites and ground stations)
Maintenance	Easier to maintain , faults are usually localized	Difficult and expensive (satellite servicing is rare and costly)

More on satellite communication :

Satellite communication systems utilize a wide range of frequencies, generally falling between 1 GHz and 40 GHz, with common bands including L, C, X, Ku, and Ka bands. These bands are used for various satellite communication applications, including mobile satellite services, fixed satellite services, TV broadcasting, and broadband internet.

Elaboration:

L-Band (1-2 GHz):

Primarily used for Mobile Satellite Service (MSS), including global positioning systems (GPS) and satellite mobile phones like Inmarsat and WorldSpace.

C-Band (4-8 GHz):

Used for Fixed Satellite Service (FSS) and TV broadcasting, particularly in areas with heavy precipitation.

X-Band (8-12 GHz):

Used for terrestrial earth exploration and meteorological satellites.

Ku-Band (12-18 GHz):

Used for a range of high-throughput communications applications, including military, emergency, and government communication systems.

Ka-Band (26-40 GHz):

Used for broadband internet services and certain military and government communication systems.